

UNIT 6

PERFORMING CALCULATIONS USING FUNDAMENTAL FUNCTIONS IN EXCEL: AVERAGE AND COUNT



Learning Objectives

- **Apply the AVERAGE function** to calculate the mean of a data range in Excel.
- **Use the COUNT function** to determine the number of numerical entries within a given range.
- **Analyze data trends** by calculating the average and counting specific entries using Excel functions.
- **Interpret numerical data** using the AVERAGE and COUNT functions for practical applications such as sales analysis or academic performance tracking.
- **Create and manage data sheets** in Excel for performing basic calculations like averages and counts.

Microsoft Excel is a powerful tool that helps us work with numbers, organize data, and quickly find patterns. Two of the most useful functions in Excel are **AVERAGE** and **COUNT**. These functions help us calculate averages and count specific entries in a list, making it easier to understand and work with large amounts of data.

1. The AVERAGE Function

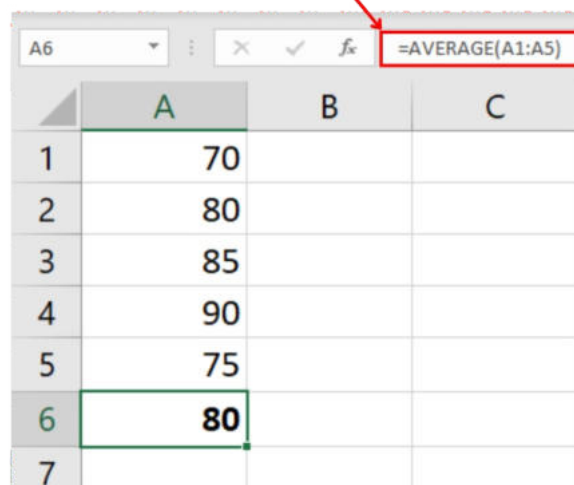
The **AVERAGE** function helps you find the **mean** of a set of numbers. To find the average, Excel adds all the numbers together and then divides the sum by the total number of numbers. This is useful when you want to find the general trend or average value of your data, like the average score of students in a class or the average temperature in a month.

How to Use the AVERAGE Function:

1. **Click on the cell** where you want to display the average.
2. Type **=AVERAGE(** and then select the range of cells you want to average. For example, if you want to find the average of numbers in cells **A1 to A5**, click and drag from **A1 to A5**.
3. After selecting the cells, close the parentheses **)** and press **Enter**.

Example:

- Suppose the scores of five students in a maths test are listed in cells **A1 to A5**: 70, 80, 85, 90, and 75. To find the average score, you would type:
 - **=AVERAGE(A1:A5)** and press **Enter**.
 - Excel will calculate and display the average score in the cell you selected.



	A	B	C
1	70		
2	80		
3	85		
4	90		
5	75		
6	80		
7			

2. The COUNT Function

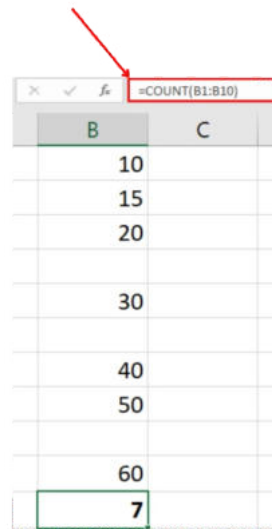
The **COUNT** function helps you count how many cells contain **numbers** in a given range. This function is very useful when you want to know how many students have submitted their homework or how many items are in a list.

How to Use the COUNT Function:

1. **Click on the cell** where you want to show the count result.
2. Type **=COUNT(** and then select the range of cells you want to count. For example, if you want to count how many numbers are in cells **B1 to B10**, click and drag from **B1 to B10**.
3. Close the parentheses **)** and press **Enter**.

Example:

- Suppose you have the following numbers in cells **B1 to B10**: 10, 15, 20, (empty), 30, (empty), 40, 50, (empty), 60.
- To count how many numbers are there, type:
- **=COUNT(B1:B10)** and press **Enter**.
- Excel will count and display the result in the selected cell, showing the total number of numbers in the range.



The screenshot shows an Excel spreadsheet with a formula bar at the top displaying `=COUNT(B1:B10)`. A red arrow points from the formula bar to the data table below. The table has two columns, B and C, and ten rows of data. Column B contains the values 10, 15, 20, 30, 40, 50, and 60. Column C is empty. The formula bar is highlighted with a red box, and a red arrow points from it to the data table.

B	C
10	
15	
20	
30	
40	
50	
60	
7	

Practical Applications of AVERAGE and COUNT Functions

1. AVERAGE:

- You can use the **AVERAGE** function to calculate the average marks of students in a class. For example, if you have marks for 10 students in cells **A1 to A10**, you can quickly find the average score using **=AVERAGE(A1:A10)**.
- It can also be used to calculate the average monthly temperature, average spending, or average time spent on homework.

2. COUNT:

- You can use the **COUNT** function to know how many students have submitted their homework or how many sales transactions were made.
- For example, if you want to know how many students scored above 50 marks in a test, you can use **COUNT** to count only the students with marks greater than 50.

Quick Tips for Using AVERAGE and COUNT:

- **AVERAGE** gives you the overall trend in your data, helping you see if things are going well or if something is too high or low.
- **COUNT** helps you know how many items, people, or events are in your dataset.
- Both functions are extremely useful when working with large datasets, such as sales reports, student grades, or inventory lists.

EXERCISES

01: Choose the correct answer.

1. Which of the following functions helps you calculate the mean of a range of numbers in Excel?

- (a) SUM
- (b) AVERAGE
- (c) COUNT

2. What does the COUNT function in Excel do?

- (a) Adds up all the numbers in a range
- (b) Counts the number of cells with numbers
- (c) Finds the highest number in a range

3. Which of these formulas will give the average of the values in cells A1 to A5?

- (a) =SUM(A1:A5)
- (b) =AVERAGE(A1:A5)
- (c) =COUNT(A1:A5)

4. What does the COUNT function ignore in a range of cells?

- (a) Empty cells
- (b) Text values
- (c) Both a and b

5. To find the average of the scores in cells B1 to B10, which formula would you use?

- (a) =COUNT(B1:B10)
- (b) =AVERAGE(B1:B10)
- (c) =SUM(B1:B10)

02: Fill in the blanks.

1. The formula for calculating the average of a range of numbers in Excel is _____.
2. The COUNT function can be used to count the number of _____ in a range of cells.
3. To use the AVERAGE function in Excel, you type _____ in a cell.
4. If you want to calculate the total number of students who have passed the exam, you would use the _____ function.
5. The formula =AVERAGE(A1:A10) calculates the average of the numbers in cells _____.

03: Tick True or False for each statement.

	TRUE	FALSE
The AVERAGE function can calculate the mean of both numbers and text in Excel.	☐	☐
The COUNT function can count both numerical and non-numerical entries in a range.	☐	☐
You can freeze more than one row or column using the Freeze Panes feature.	☐	☐
You can use the AVERAGE function to find the average of a set of student scores.	☐	☐
The COUNT function ignores empty cells in the range.	☐	☐
The AVERAGE function divides the sum of values by the number of entries in the selected range.	☐	☐

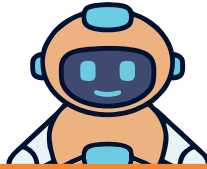
04: Answer the following questions briefly.

- 1.** What is the purpose of the AVERAGE function in Excel? Give one real-life example where you can use it.
- 2.** How does the COUNT function differ from the AVERAGE function?
- 3.** How can you find out how many students in a class have scored above 50 marks using Excel?
- 4.** If you want to calculate the average test scores of 20 students in a class, which function will you use and why?
- 5.** How would you use the COUNT function to know how many responses were received for a survey?

05: Provide detailed answers for the following questions.

- 1.** Explain how you would use both the AVERAGE and COUNT functions in Excel to analyze the results of a class test. Include step-by-step instructions on how to calculate the average score and count the number of students who passed the test.
- 2.** Suppose you are analyzing monthly sales data for a shop. How would you use the AVERAGE and COUNT functions to summarize the sales performance? Include examples of how to apply both functions to calculate the average sales and count the number of months with high sales.
- 3.** You have the sales data of 10 products, and you want to find out the total number of products that have been sold more than 50 times. How would you apply the COUNT function to solve this problem? Explain with an example.
- 4.** How would you use the COUNT and AVERAGE functions to assess the performance of students in a subject? Explain the steps and the importance of both functions in evaluating their results.

01: Lab Activity : Counting Data in Excel



To practise using the COUNT formula in Excel.

Instructions:

1. Open a new Excel workbook.
2. Create at least three sheets and name them according to different topics (e.g., "Maths", "Science", "English").
3. In each sheet, input some data (e.g., numbers of students, scores, etc.).
4. Use the COUNT formula to count the number of cells that contain numbers in each sheet. Example: **=COUNT(A1:A10)**
5. Save the workbook for future use.

02: Lab Activity : Calculating the Average of exam scores



To practise using the AVERAGE formula in Excel.

Instructions:

1. Open a new Excel workbook.
2. Create at least three sheets and name them according to different topics (e.g., "Maths", "Science", "English").
3. In each sheet, input some data (e.g., numbers of students, scores, etc.).
4. Use the AVERAGE formula to count the number of cells that contain numbers in each sheet. Example: **=AVERAGE(A1:A10)**
5. Save the workbook for future use.